

Scripts for Describing Simple Charts and Graphs in Under 120 Words.

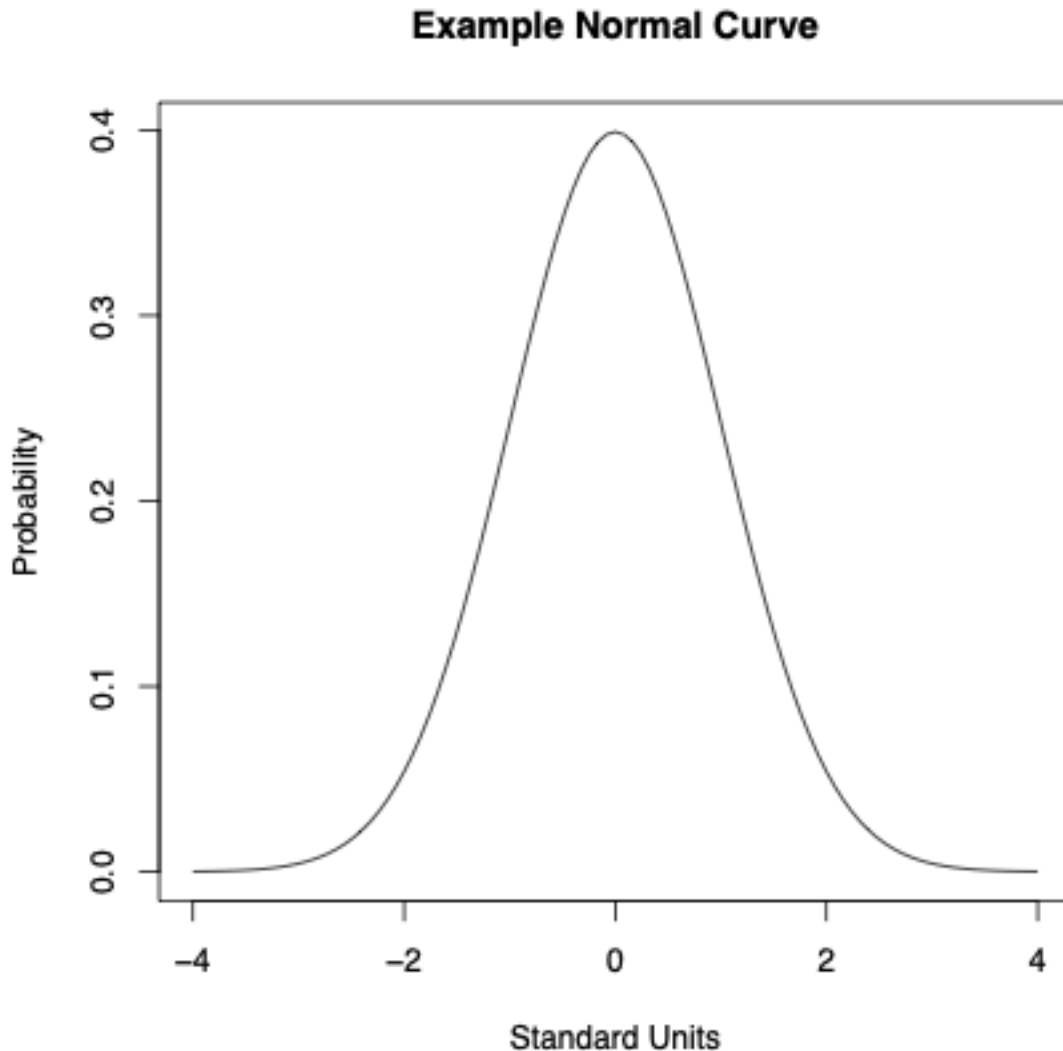
This document contains written scripts to describe following charts and graphs that are regularly used in introductory statistics.

The scripts were influenced by descriptions generated for some types of graphs using the VI function within the BrailleR package in R developed by Godfrey and colleagues (2023). The text that is specific to a given plot is shown in bold. It is intended that the main trends present in the plot will be described in the main text body, while the script to describe the axis and element information will be written as alternative text. For simple graphs and charts, as shown below, a complete description is possible within a 120-word limit.

All data in these plots are fictional and have been created for demonstration purposes only.

1. Smooth Curve
 - a. Left-right symmetrical
 - b. Skewed
2. Histogram
3. Boxplot
 - a. Horizontal
 - b. Vertical
4. Bar chart
 - a. In alphabetical order
 - b. In descending order
5. Pie Chart
6. Dot plot
 - a. Horizontal
 - b. Vertical
7. Line graph
8. Scatterplot
9. Continuous linear regression plot
10. Logistic regression plot

1. Smooth curve



A. A normal curve that is left-right symmetric about the x axis.

Smooth normal curve of: **Example Normal Curve**.

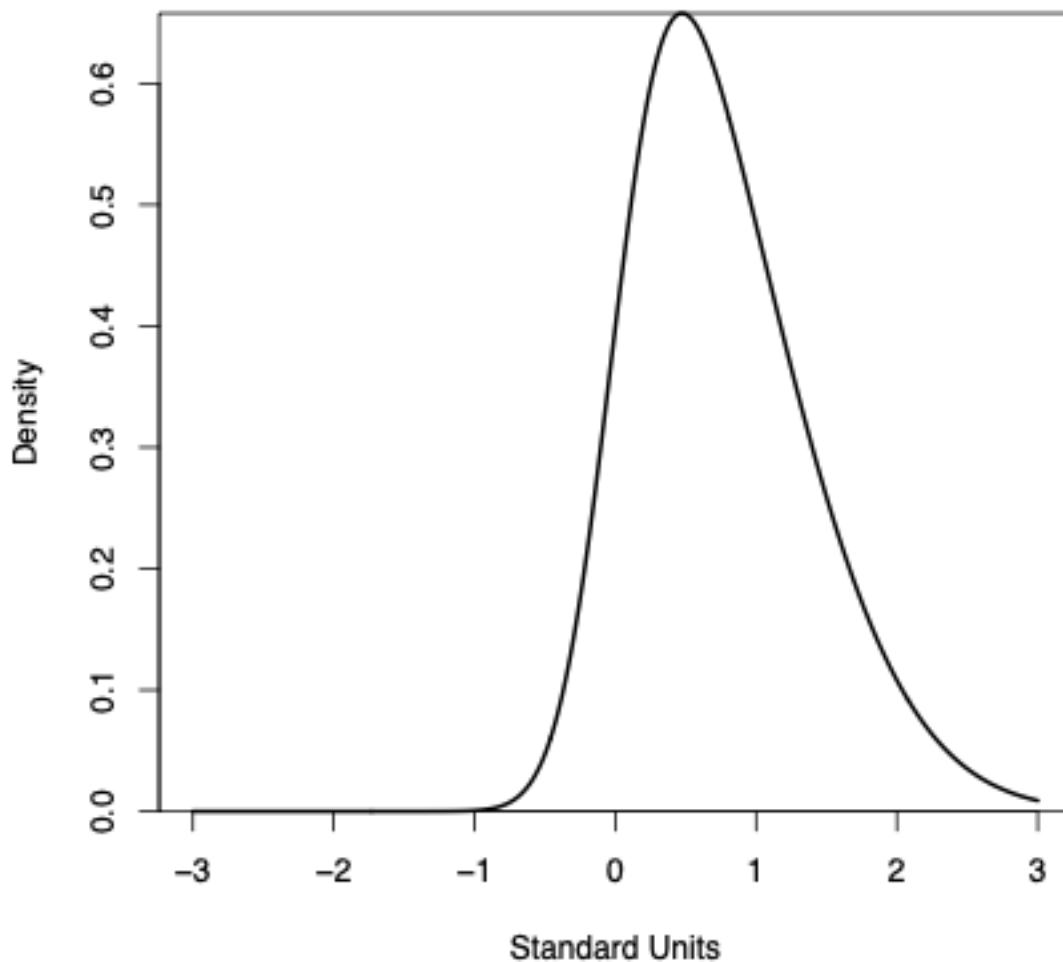
The x-axis is **Standard Units**, from **-4 to 4** and the y-axis is **Probability**, from **0 to 0.4**.

Tick marks on the x-axis from left to right are at: **-4, -2, 0, 2, and 4**.

Tick marks on the y-axis from bottom to top are at: **0, 0.1, 0.2, 0.3, and 0.4**.

The mean is **0** and the standard deviation is **1**.

Example Skewed Curve



B. A left-skewed curve with a longer tail to the left and most values concentrated toward the right.

Left-skewed smooth curve of: **Example Skewed Curve**

The x-axis is **Standard Units**, from **-3 to 3** and the y-axis is **Density**, from **0 to 0.6**.

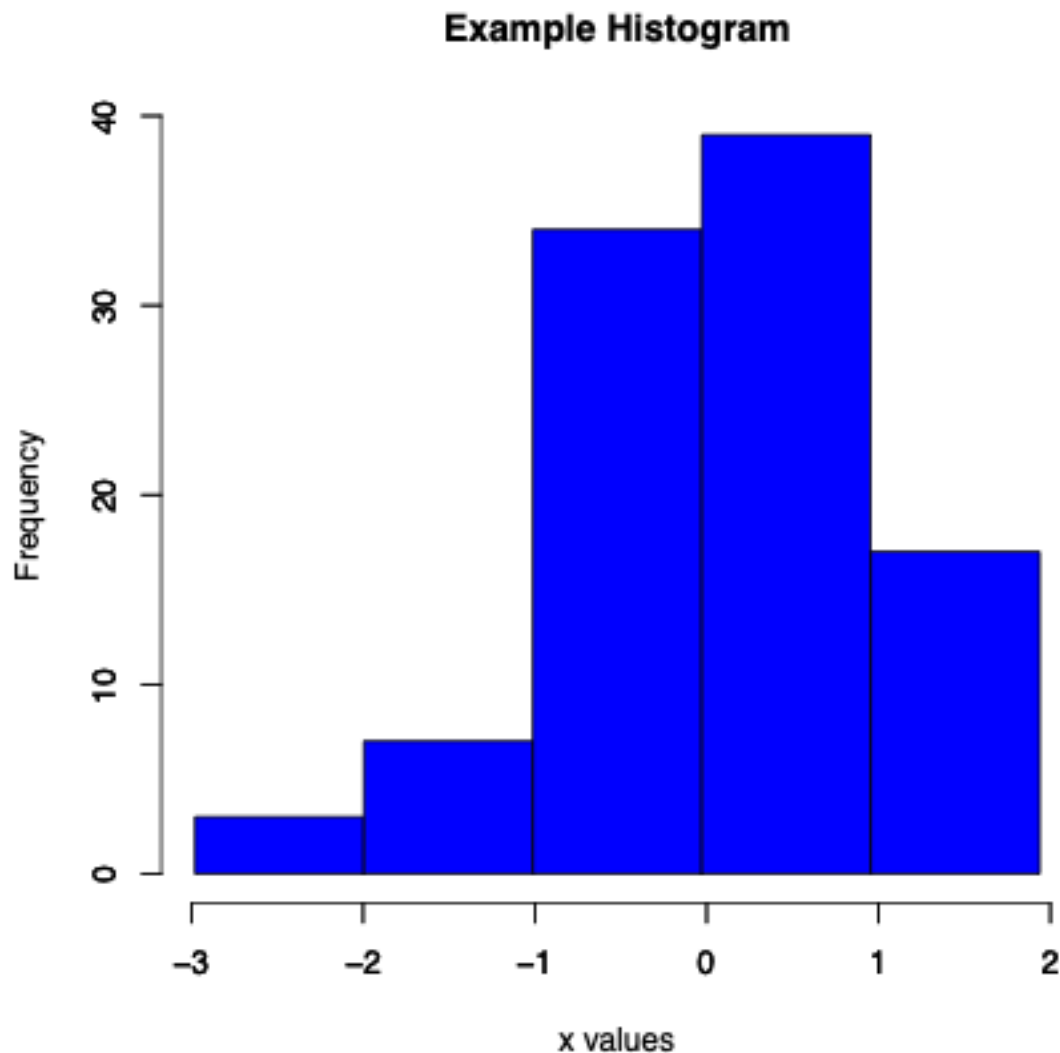
Tick marks on the x-axis from left to right are at: **-3, -2, -1, 0, 1, 2, and 3**.

Tick marks on the y-axis from bottom to top are at: **0, 0.1, 0.2, 0.3, and 0.4, 0.5, and 0.6**.

The mean is **0** and the standard deviation is **1**.

The alpha, or skew, parameter is **3**.

2. Histogram



The histogram shows a **slightly left skewed normal distribution of values with a mean of 0 and a standard deviation of 1.**

Histogram of an **Example Distribution.**

The x-axis is **x values in standard units, from -3 to 2** and the y-axis is **Frequency, from 0 to 39.**

There are **100** observations.

Tick marks on the x-axis are at: **-3, -2, -1, 0, 1, and 2.**

Tick marks on the y-axis are at: **0, 10, 30, and 40.**

5 intervals with equal widths.

For interval **-3 to -2**, the count is **3.**

For interval **-2 to -1**, the count is **7.**

For interval **-1 to 0**, the count is **34.**

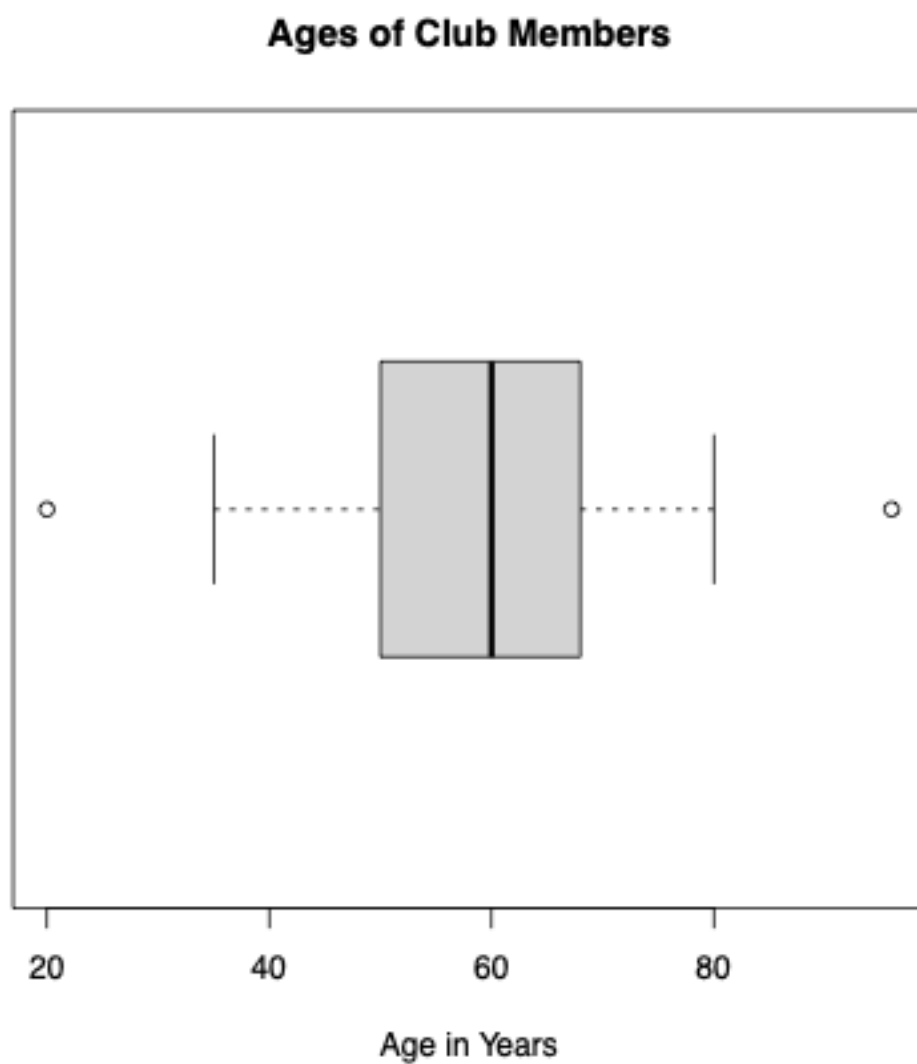
For interval **0 to 1**, the count is **39.**

For interval **1 to 2**, the count is **17.**

3.Boxplot

Table 1: Comparison of quartiles and quantiles with percentiles

Quartile	Quantile	Percentile
0	0	0
1	0.25	25
2	.5	50
3	.75	75
4	1	100



- A.** The horizontal box plot shows a **distribution of the Ages of Club Members**, mostly between 50 and 68 years, and centered at 60 years. There is one unusually young member at 20 years, and one unusual old member at 96 years.

Horizontal boxplot of: Ages of Club Members.

The distribution of **age in years**.

There are **70** observations.

The Q0 is **20**.

The Q1 is **50**.

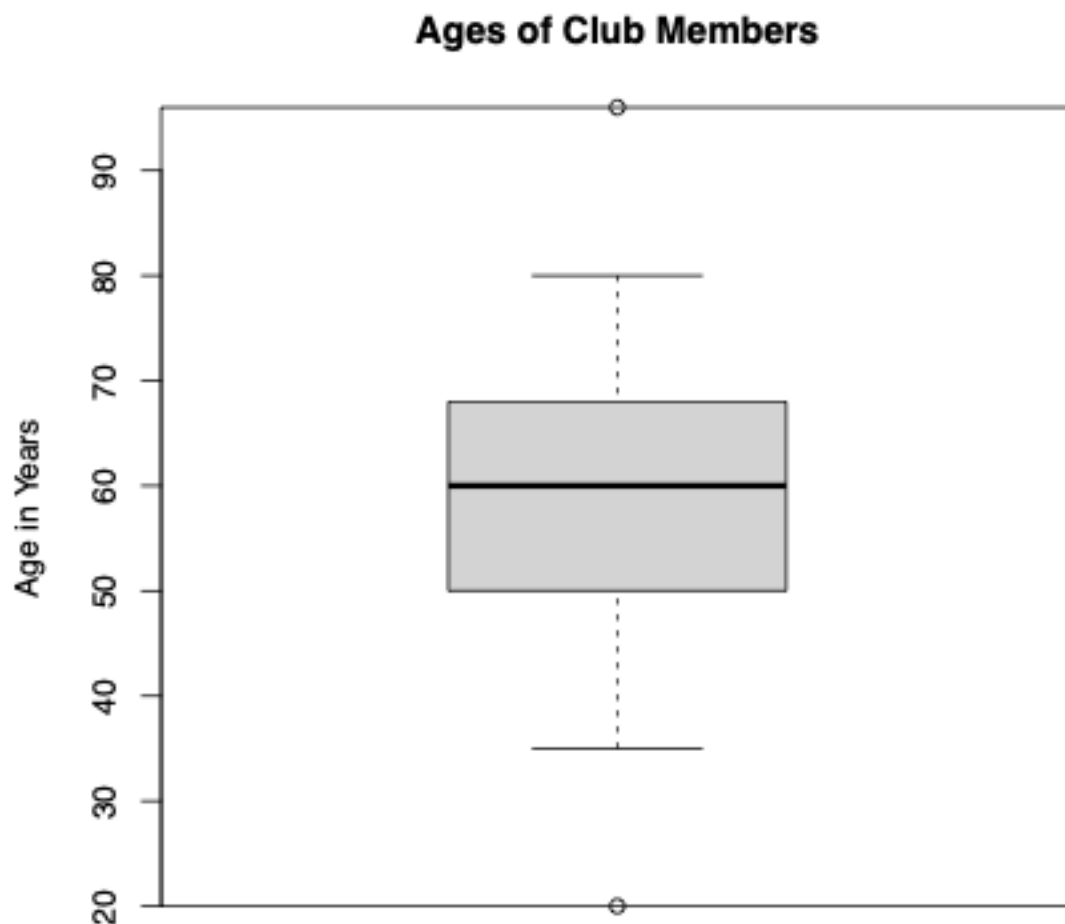
The Q2 and median is **60**.

The Q3 is **68**.

The Q4 is **96**.

The small value fence is **35** and the large value fence is **80**.

There is **1** small outlier of **20** and **1** large outlier of **96**.



- B.** The vertical box plot shows a distribution of the Ages of Club Members, mostly between 50 and 68 years, and centered at 60 years. There is one unusually young member at 20 years, and one unusual old member at 96 years.

Vertical boxplot of: Ages of Club Members.

The distribution of **age in years**.

There are **70** observations.

The Q0 is **20**.

The Q1 is **50**.

The Q2 and median is **60**.

The Q3 is **68**.

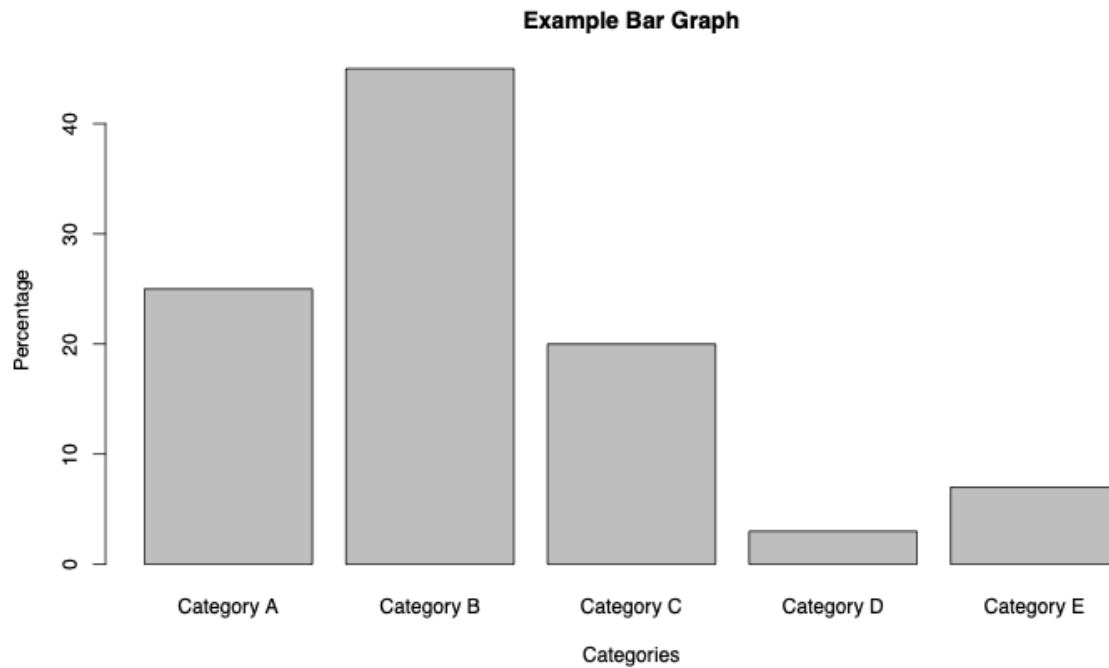
The Q4 is **96**.

The small value fence is **35** and the large value fence is **80**.

There is **1** small outlier of **20** and **1** large outlier of **96**.

4. Bar chart

A. Categories in alphabetical order



A. The bar graph shows **the percentages of 5 categories, from A to E, in alphabetical order. Category B has by far the greatest percentage, of around 45.**

Bar chart of **an Example Bar Graph.**

The x-axis is **Categories**, and the y-axis is **Percentages**, from **3** to **45**.

Labels of the x-axis are: **Category A, Category B, Category C, Category D, and Category E.**

Tick marks on the y-axis are at: **0, 10, 20, 30, and 40**

5 bars with labels and lengths:

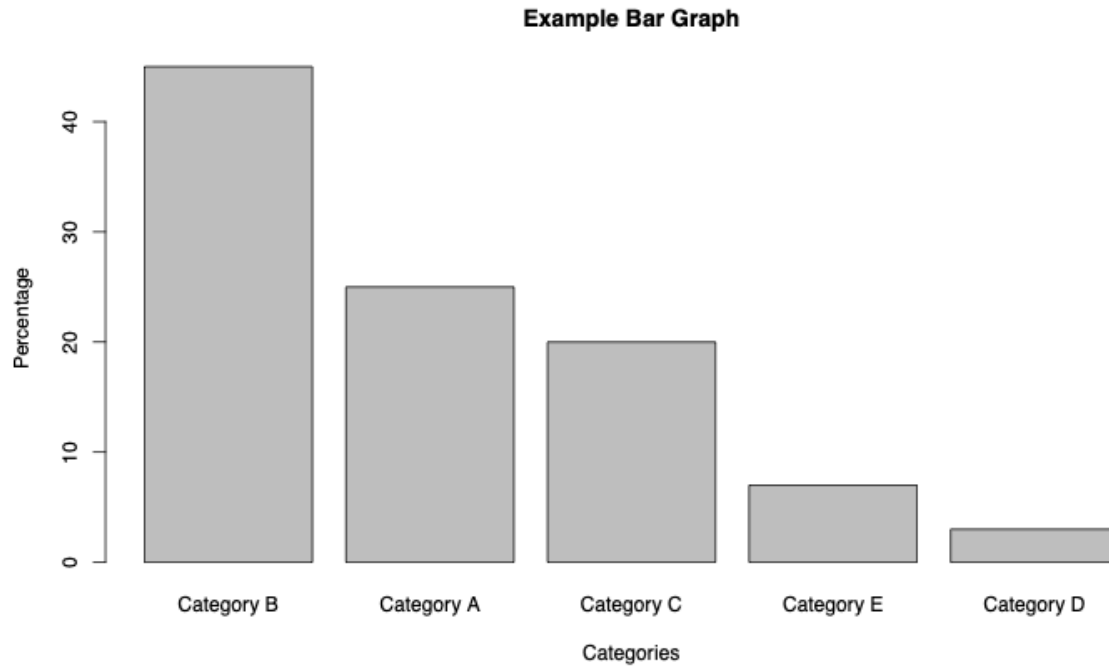
Category A, from **0** to **25**

Category B, from **0** to **45**

Category C, from **0** to **20**

Category D, from **0** to **3**

Category E, from **0** to **7**



B. The bar graph shows the percentages of 5 categories, from A to E, in descending order. Category B has by far the greatest percentage, of around 45.

Bar chart of an **Example Bar Graph**.

The x-axis is **Categories**, and the y-axis is **Percentages**, from 3 to 45.

Labels of the x-axis are: **Category B**, **Category A**, **Category C**, **Category E**, and **Category D**.

Tick marks on the y-axis are at: **0**, **10**, **20**, **30**, and **40**

5 bars with labels and lengths:

Category B, from **0** to **45**

Category A, from **0** to **25**

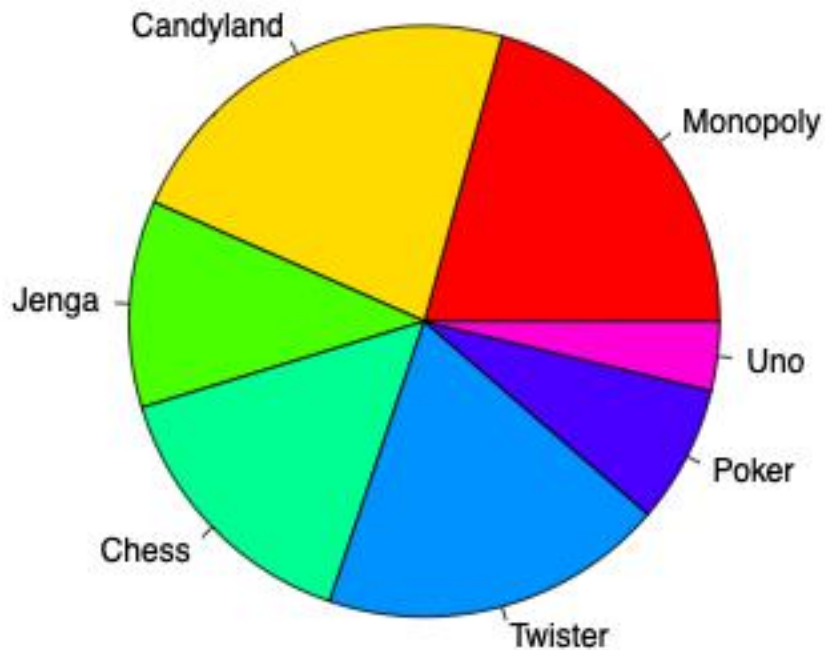
Category C, from **0** to **20**

Category E, from **0** to **7**

Category D, from **0** to **3**

5. Pie chart

Percentage of Players by Game Type



The pie chart shows **percentages of players per game type**, with **Candyland having the most players** and **Uno having the fewest**.

Pie chart of: **Percentage of Players by Game Type**.

There are **53 players**.

7 categories are scaled to their percentage size.

Candyland has **22.6%**.

Monopoly has **20.8%**.

Twister has **18.9%**.

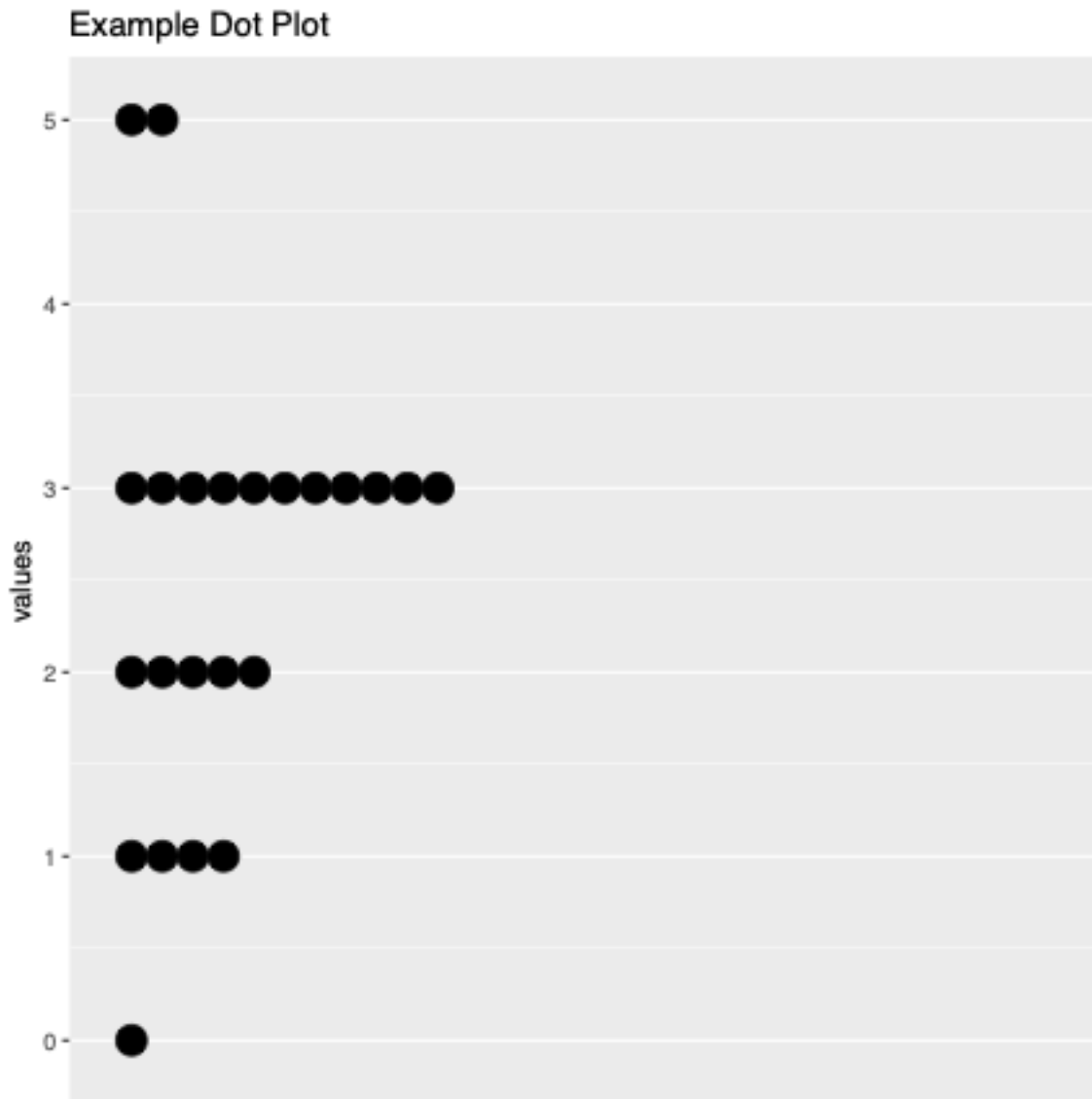
Chess has **15.1%**.

Jenga has **11.3%**.

Poker has **7.5%**.

Uno has **3.8%**.

6. Dot plot or dot chart



- A. A horizontal dot plot shows **values that range between 0 and 5, building up to 3 and then dropping steeply off.**

Horizontal dot plot of: Example Dot Plot.

The x-axis is unlabeled but shows the **percentage of the count** of each **value from 0% to 100%**, with counts from **0 to 11**, and the y-axis is **values**, from **0 to 5**.

There are no tick marks on the x-axis.

Tick marks on the y-axis are at: **0, 1, 2, 3, 4, and 5.**

There are **23** observations.

6 marks with stacked horizontal dots.

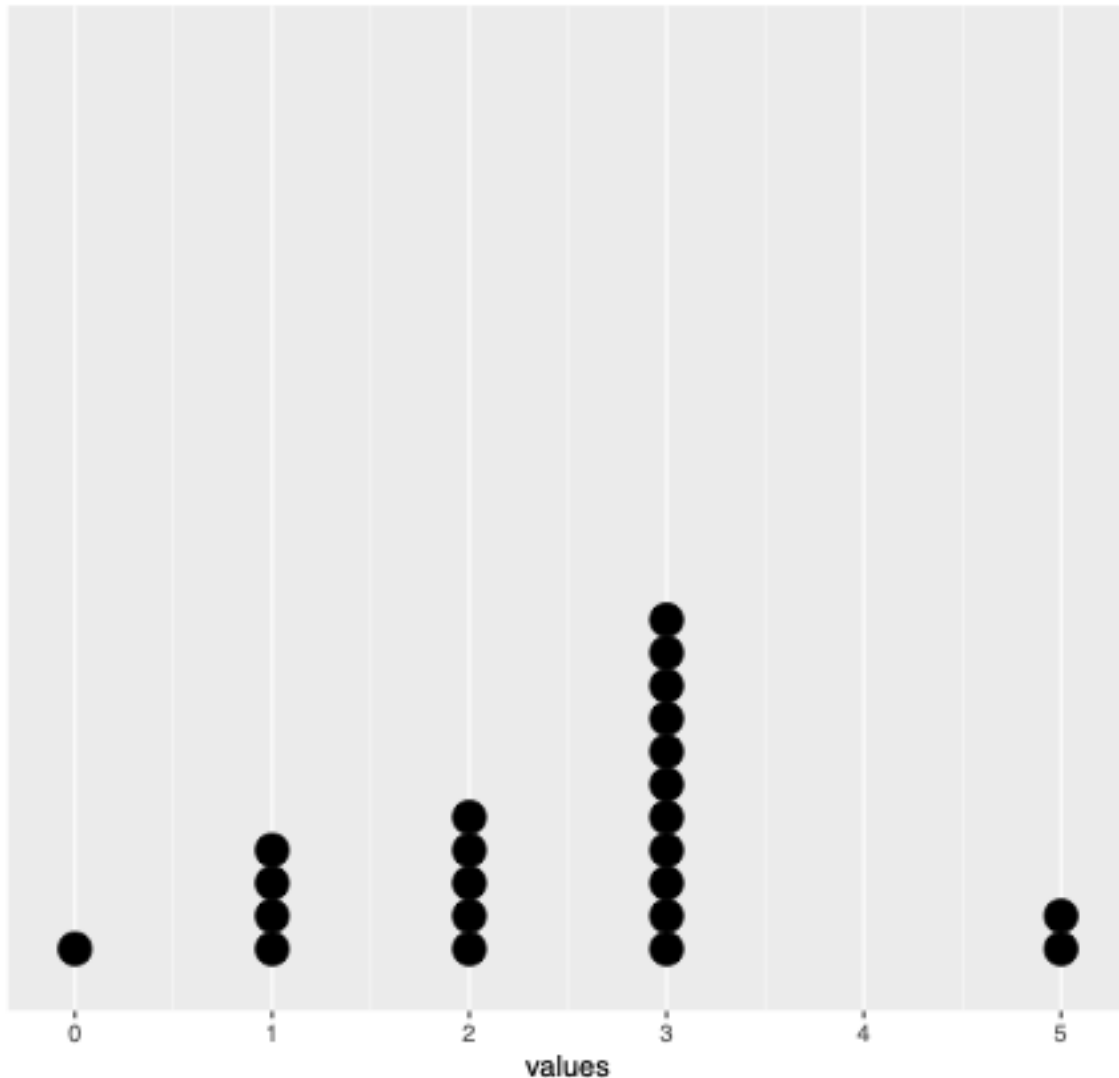
At mark **0**, there is **1 dot**.

At mark **1**, there are **4 dots**.

At mark **2**, there are **5 dots**.

At mark **3**, there are **11 dots**.
At mark **4**, there are **0 dots**.
At mark **5**, there are **2 dots**.

Example Dot Plot



B. A vertical dot plot shows **values** that range between 0 and 5, building up to 3 and then dropping steeply off.

Vertical dot plot of: **Example Dot Plot**.

The x-axis is **values**, from **0 to 5** and the y-axis is unlabeled but shows the **percentage of the count** of each **value from 0% to 100%**, with counts from **0 to 11**. Tick marks on the x-axis are at: **0, 1, 2, 3, 4, and 5**.

There are no tick marks on the y-axis.

There are **23** observations.

6 marks with stacked vertical dots.

At mark **0**, there is **1 dot**.

At mark **1**, there are **4 dots**.

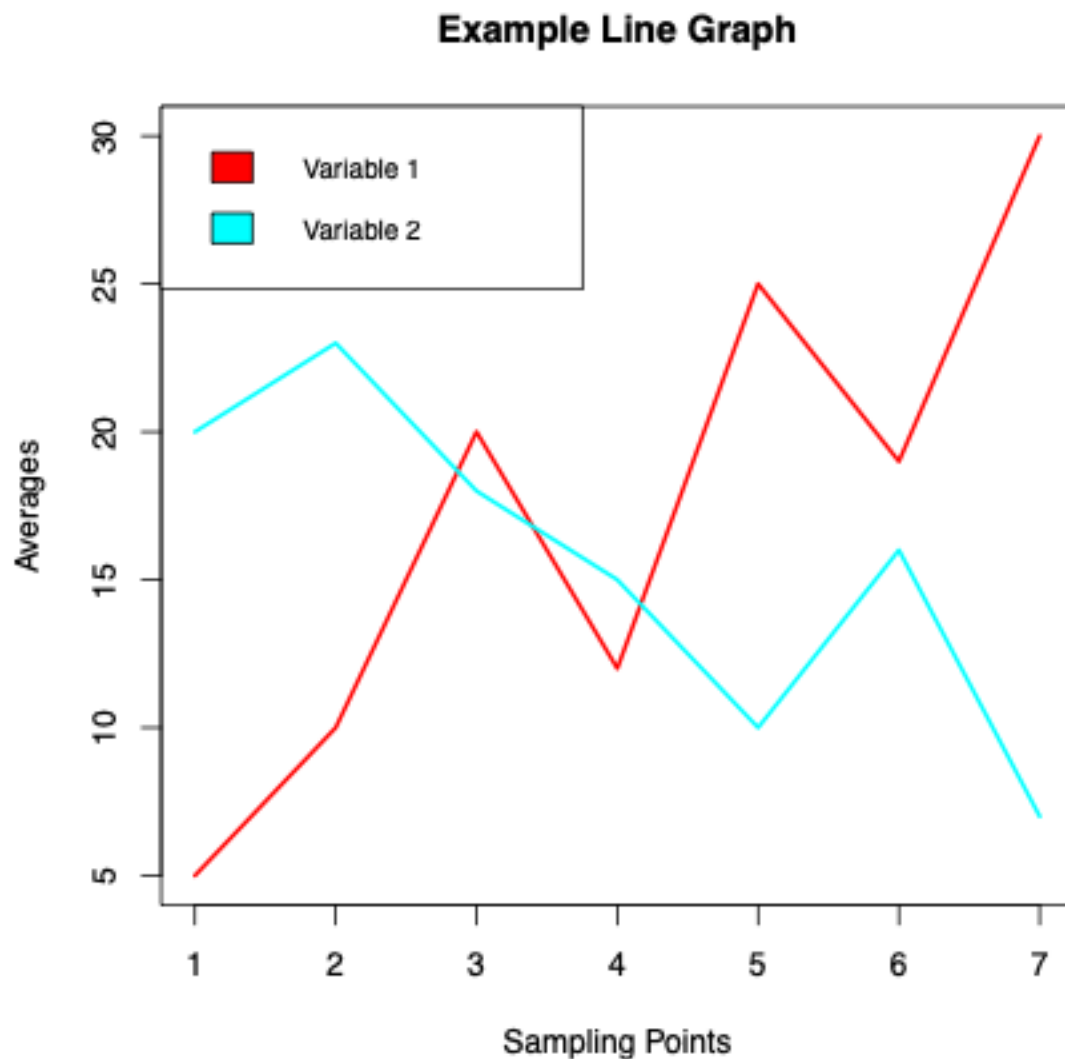
At mark **2**, there are **5 dots**.

At mark **3**, there are **11 dots**.

At mark **4**, there are **0 dots**.

At mark **5**, there are **2 dots**.

7.Line graph



The line graph shows **average values** for two variables at **sampling points 1 through 7**. **Variable 1** is a red line that generally moves upward and **Variable 2** is an aquamarine-colored line that generally moves downward, as the **sampling points increase**. There is variability in the values such that the connecting lines make a jagged rather than a straight pattern.

Line graph of: **Example Line Graph**.

The x-axis is **Sampling Points**, from **1 to 7**, and the y-axis is **Averages**, from **5 to 30**.

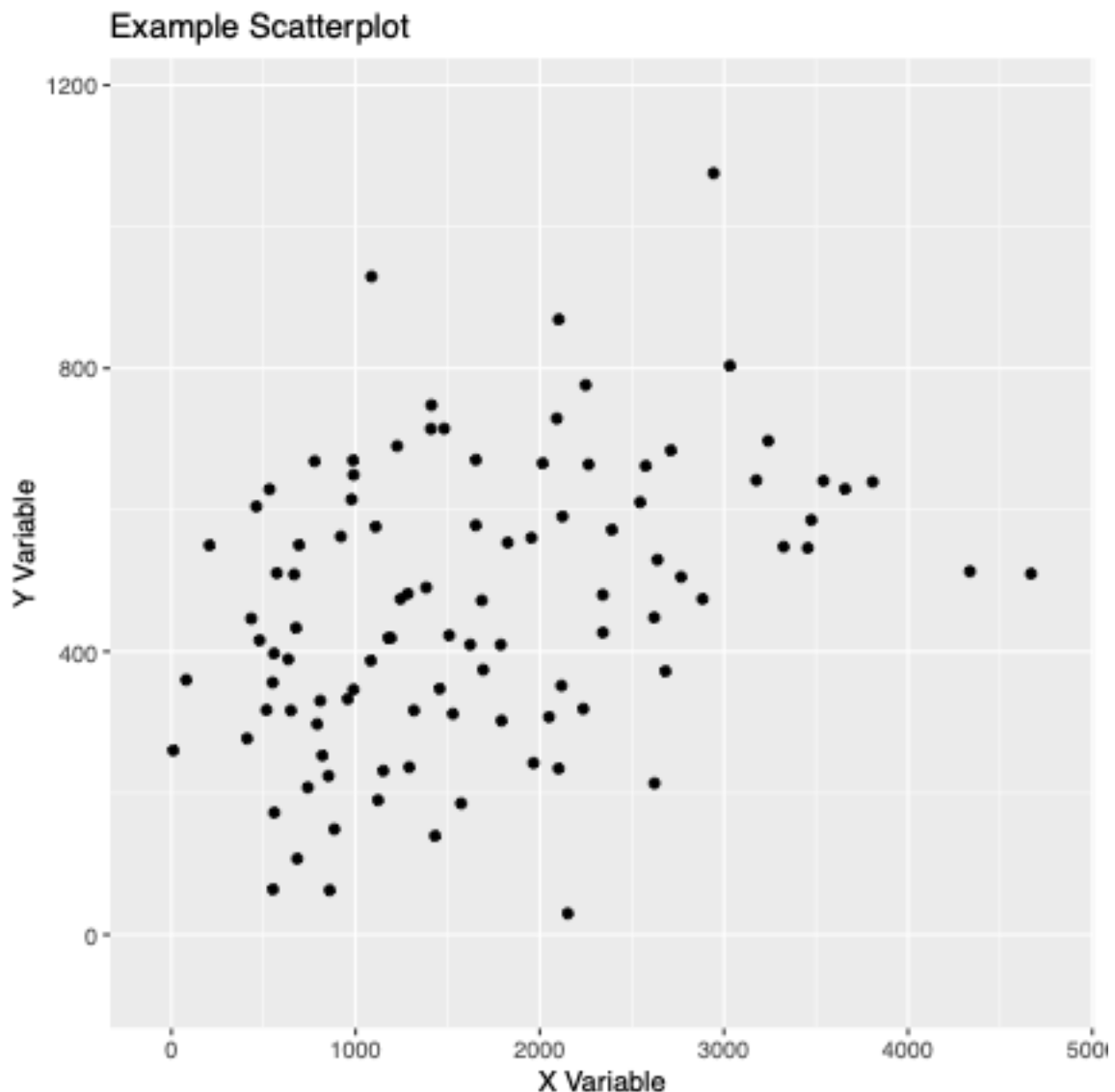
Tick marks on the x-axis are at: **1, 2, 3, 4, 5, 6, and 7**.

Tick marks on the y-axis are at: **5, 10, 15, 20, 25, and 30**.

Variable 1, a red line, connects **7** mean values at the **7** x-axis tick marks. The mean values are: **5, 10, 20, 12, 25, 19, and 30**.

Variable 2, an **aqua colored** line, connects **7** mean values at the **7** x-axis tick marks. The values are: **20, 23, 18, 15, 10, 16**, and **7**.

8. Scatterplot



The scatter plot shows a **positive relationship** between the **X Variable** that **ranges between 10 and 4668 units**, and the **Y Variable** that **ranges between 30 and 1076 units**, with a **correlational coefficient of around 0.36**.

Scatterplot of: **Example Scatterplot**.

The x-axis is **X Variable with unspecified units**, from **10 to 4668 units** and the y-axis is **Y Variable with unspecified units**, from **30 to 1076 units**.

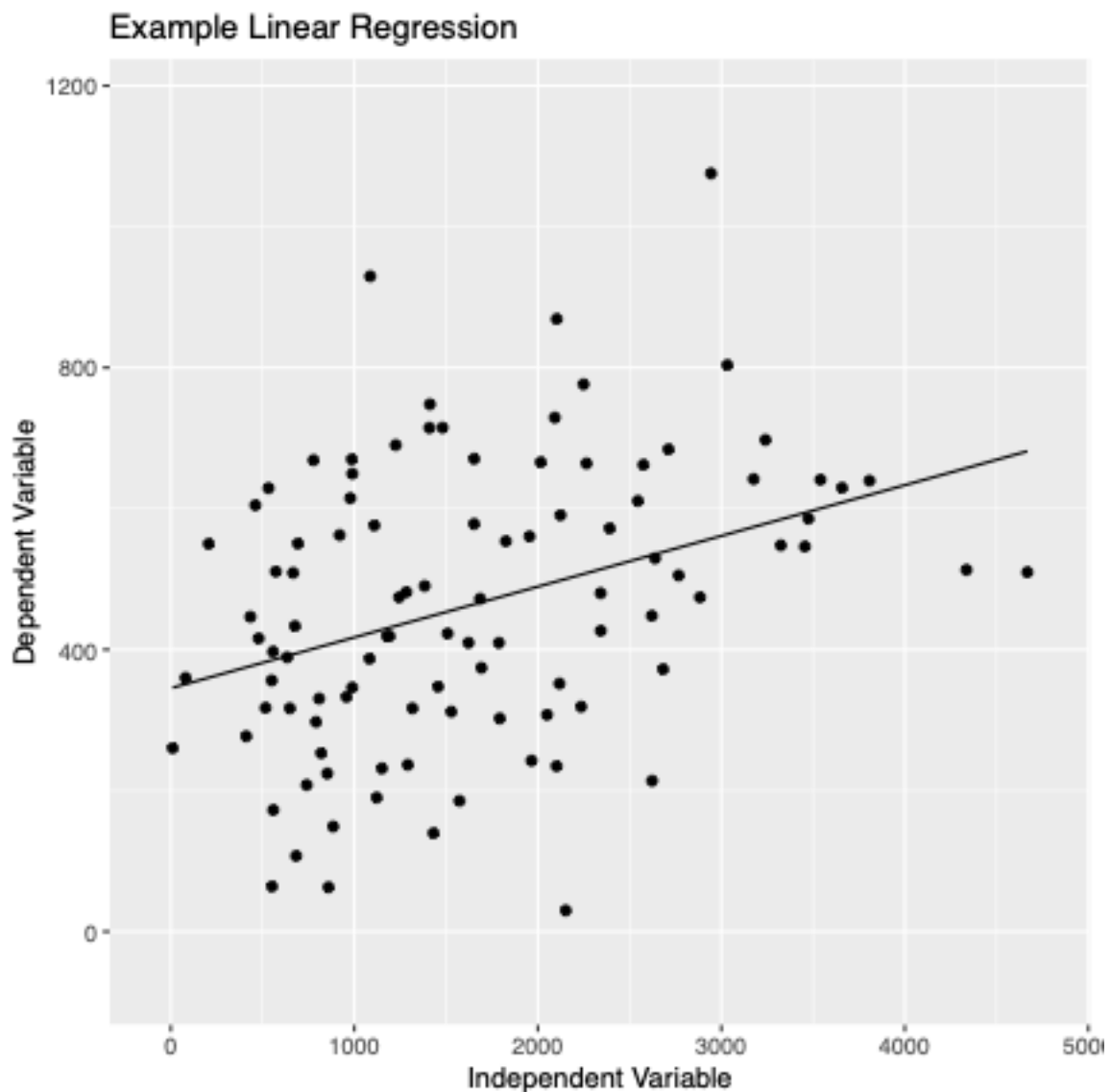
Tick marks on the x-axis are at: **0, 1000, 2000, 3000, 4000, and 5000**.

Tick marks on the y-axis are at: **0, 400, 800, and 1200**.

There are **100** observations.

The correlational coefficient is **0.36**.

9. Linear Regression Plot



The linear regression plot shows a **positive relationship between the Independent Variable that ranges between 10 and 4668 units, and the Dependent Variable that ranges between 30 and 1076 units, with a correlational coefficient of around 0.36.**

Continuous linear regression plot of: **Example Linear Regression.**

The x-axis is the **Independent Variable with unspecified units, from 10 to 4668 units** and the y-axis is the **Dependent Variable with unspecified units, from 30 to 1076 units.**

Tick marks on the x-axis are at: **0, 1000, 2000, 3000, 4000, and 5000.**

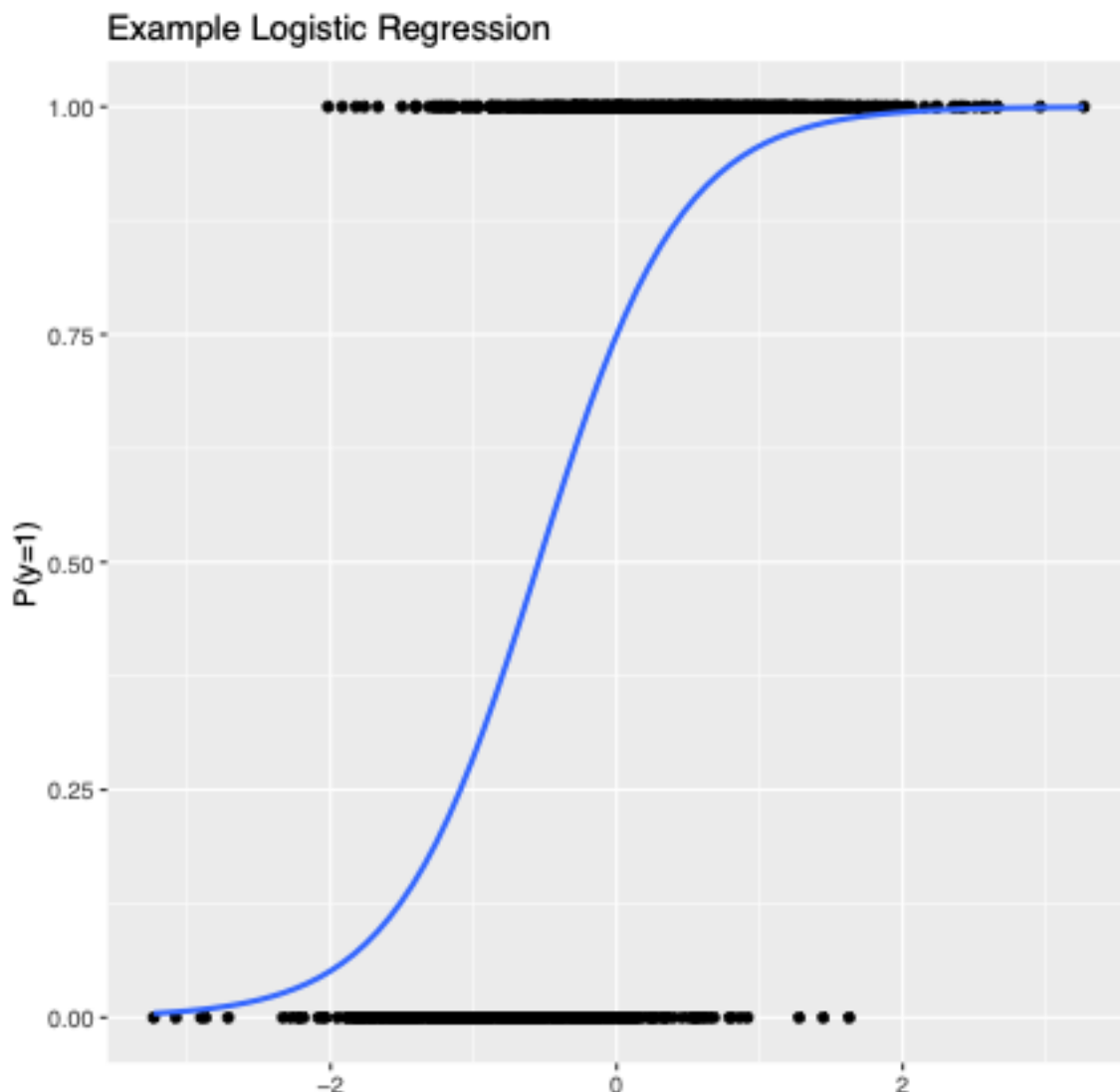
Tick marks on the y-axis are at: **0, 400, 800, and 1200.**

There are **100** observations.

The correlational coefficient is **0.36.**

The slope of the regression line is estimated as **0.072**, and the y-intercept is estimated as **345.**

10. Logit Regression Plot



The logistic regression plot with a **binomial outcome variable of 0 or 1**, shows a **decision boundary at around -0.6** in the feature dimension.

Logistic regression plot of: **Example Logistic Regression.**

The x-axis is an **unlabeled feature dimension ranging from around -3.5 to around 3.5**, and the y-axis is the **probability of $y = 1$** , from 0 to 1, with a **decision boundary at 0.5**.

Tick marks on the x-axis are at: **-2, 0, and 2.**

Tick marks on the y-axis are at: **0, 0.25, 0.5, 0.75, and 1.**

There are **1000** observations.

For $y = 0$, values range from **-3.2 to 1.6.**

For $y = 1$, values range from **-2.0 to 3.3.**

A smooth sigmoid function is shown in **blue**.

The decision boundary falls **-0.6 in the feature dimension**.

References

Godfrey, A. J. R. (2023). *BrailleR: Improved Access for Blind Users* (Version 1.0.2) [Computer software]. CRAN (archived). <https://cran.r-project.org/package=BrailleR>